

1. Model Description

Parameter	Value
Base model	AML_AE26_ChainDrive__04_spring_update_redMass.etd
Gap variant model	AML_AE26_ChainDrive__04_spring_update_redMass_HLA_Var.etd
Base speed cases	7300 rpm, 7400 rpm, 7500 rpm, 7600 rpm, 7700 rpm, 7800 rpm
Gap variant speeds	7500, 7600, 7700, 7800 rpm
HLA gap values	0.0053 mm (tight) and 0.09 mm (loose)
Simulation duration	5400 deg cam / 15 cam cycles total
Analysis window	Last cam cycle (base model); last 5 cycles (gap study)
Intake HLA elements	16 (INTL_HLIF1-8, INTr_HLIF1-8)
Pump-up threshold	0.1 mm (min lift on base circle, last cycle)
Software	AVL EXCITE Timing Drive R2024.1
Analysis date	2026-07-07

This report consolidates all HLA pump-up evaluations for the **AML AE26 ChainDrive 04_spring_update** intake valvetrain. Section 2 covers the **_redMass** base model across the full 7300-7800 rpm sweep. Section 3 presents the **gap sensitivity parametric study** comparing a tight gap (0.0053 mm, close to nominal) against a loose gap (0.09 mm, 17x nominal) across 7500-7800 rpm. The two sections share the same analysis methodology and visual style to facilitate direct comparison.

2. Base Model - Intake HLA Lift Pump-Up Verification

All cam cycles are overlaid on a single 0-360 deg axis for each speed point. In the absence of pump-up, successive cycles are indistinguishable and the HLA plunger returns to zero (base circle) between lift events. A non-zero or rising baseline across cycles is the primary pump-up indicator; the dashed red line marks the 0.10 mm detection threshold.

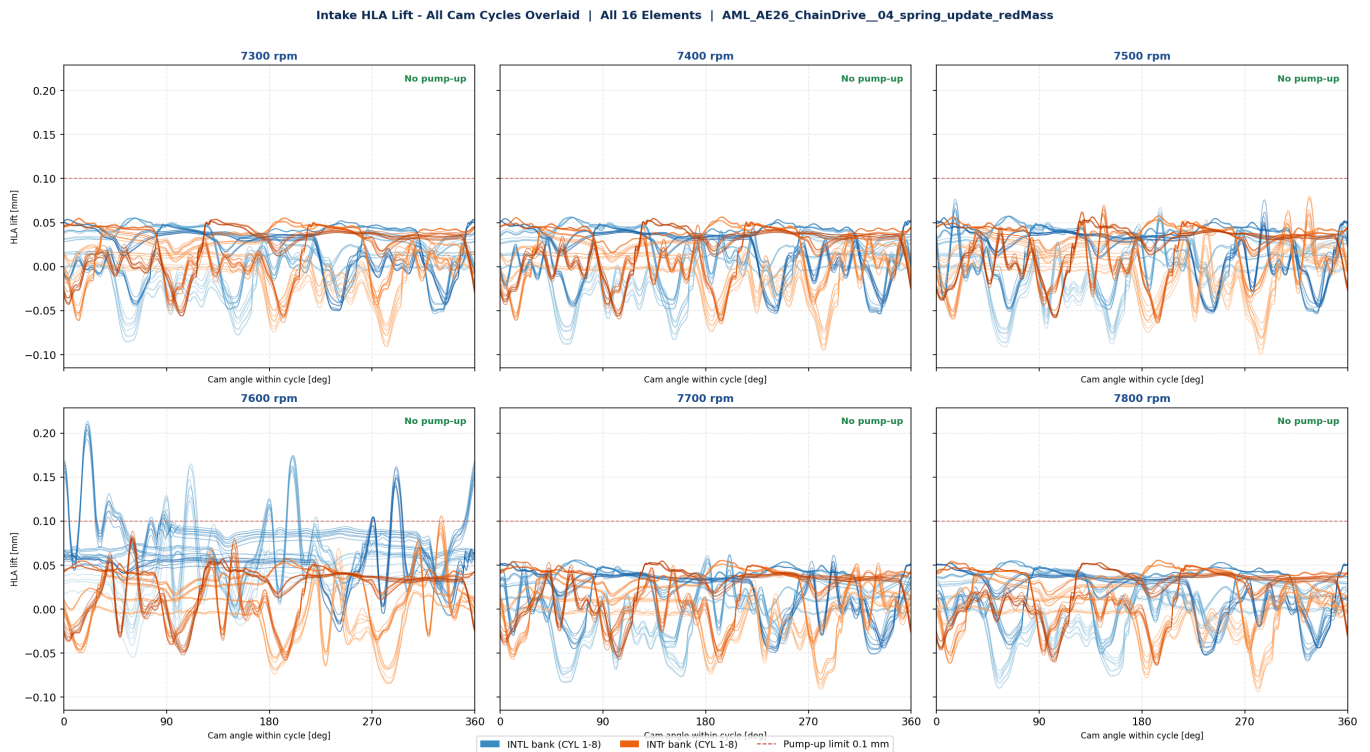


Fig. 2-1: HLA lift - all cycles overlaid, all 16 intake elements, base model. Blue = INTL bank; orange = INTr bank. Each speed in its own subplot. Cycles overlay exactly with no baseline drift, confirming zero pump-up at all speeds.



Fig. 2-2: HLA lift - last cycle, all speeds, individual element view, base model. Each cell is one intake valve. Speed-dependent variation in peak lift is small; base-circle lift remains at zero across all cylinders and speeds.

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AML AE26 ChainDrive 04 spring_update_redMass | HLA Study | 2026-07-07

2.1 HLA Pump-Up Summary - Base Model

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?
INTL_HLIF1	7300	0.0145	-0.0844	no
INTL_HLIF2	7300	0.0322	-0.0711	no
INTL_HLIF3	7300	0.0550	-0.0473	no
INTL_HLIF4	7300	0.0547	-0.0353	no
INTL_HLIF5	7300	0.0520	-0.0497	no
INTL_HLIF6	7300	0.0490	-0.0504	no
INTL_HLIF7	7300	0.0538	-0.0465	no
INTL_HLIF8	7300	0.0511	-0.0494	no
INTR_HLIF1	7300	0.0211	-0.0897	no
INTR_HLIF2	7300	0.0347	-0.0733	no
INTR_HLIF3	7300	0.0512	-0.0630	no
INTR_HLIF4	7300	0.0553	-0.0549	no
INTR_HLIF5	7300	0.0547	-0.0507	no
INTR_HLIF6	7300	0.0495	-0.0616	no
INTR_HLIF7	7300	0.0537	-0.0566	no
INTR_HLIF8	7300	0.0522	-0.0535	no
INTL_HLIF1	7400	0.0126	-0.0847	no
INTL_HLIF2	7400	0.0304	-0.0722	no
INTL_HLIF3	7400	0.0566	-0.0427	no
INTL_HLIF4	7400	0.0564	-0.0429	no
INTL_HLIF5	7400	0.0534	-0.0491	no
INTL_HLIF6	7400	0.0500	-0.0549	no
INTL_HLIF7	7400	0.0543	-0.0541	no
INTL_HLIF8	7400	0.0518	-0.0536	no
INTR_HLIF1	7400	0.0215	-0.0843	no
INTR_HLIF2	7400	0.0354	-0.0741	no
INTR_HLIF3	7400	0.0524	-0.0604	no
INTR_HLIF4	7400	0.0567	-0.0489	no
INTR_HLIF5	7400	0.0557	-0.0603	no
INTR_HLIF6	7400	0.0499	-0.0604	no
INTR_HLIF7	7400	0.0555	-0.0456	no
INTR_HLIF8	7400	0.0527	-0.0536	no
INTL_HLIF1	7500	0.0138	-0.0909	no
INTL_HLIF2	7500	0.0310	-0.0749	no
INTL_HLIF3	7500	0.0566	-0.0483	no
INTL_HLIF4	7500	0.0589	-0.0401	no

AML Valvetrain Engineering - HLA Pump-Up Analysis

AML AE26 ChainDrive 04 spring_update_redMass | HLA Study | 2026-07-07

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?
INTL_HLIF5	7500	0.0647	-0.0469	no
INTL_HLIF6	7500	0.0600	-0.0529	no
INTL_HLIF7	7500	0.0608	-0.0473	no
INTL_HLIF8	7500	0.0565	-0.0450	no
INTR_HLIF1	7500	0.0196	-0.0953	no
INTR_HLIF2	7500	0.0467	-0.0754	no
INTR_HLIF3	7500	0.0601	-0.0495	no
INTR_HLIF4	7500	0.0573	-0.0562	no
INTR_HLIF5	7500	0.0662	-0.0607	no
INTR_HLIF6	7500	0.0685	-0.0629	no
INTR_HLIF7	7500	0.0571	-0.0577	no
INTR_HLIF8	7500	0.0537	-0.0511	no
INTL_HLIF1	7600	0.1304	-0.0329	no
INTL_HLIF2	7600	0.1488	-0.0007	no
INTL_HLIF3	7600	0.1782	0.0035	no
INTL_HLIF4	7600	0.1874	0.0046	no
INTL_HLIF5	7600	0.1902	0.0111	no
INTL_HLIF6	7600	0.1828	0.0070	no
INTL_HLIF7	7600	0.1527	-0.0081	no
INTL_HLIF8	7600	0.1596	-0.0101	no
INTR_HLIF1	7600	0.0271	-0.0890	no
INTR_HLIF2	7600	0.0538	-0.0690	no
INTR_HLIF3	7600	0.1073	-0.0435	no
INTR_HLIF4	7600	0.0920	-0.0436	no
INTR_HLIF5	7600	0.0747	-0.0539	no
INTR_HLIF6	7600	0.0680	-0.0586	no
INTR_HLIF7	7600	0.0884	-0.0442	no
INTR_HLIF8	7600	0.0885	-0.0497	no
INTL_HLIF1	7700	0.0166	-0.0814	no
INTL_HLIF2	7700	0.0280	-0.0730	no
INTL_HLIF3	7700	0.0579	-0.0377	no
INTL_HLIF4	7700	0.0675	-0.0434	no
INTL_HLIF5	7700	0.0533	-0.0454	no
INTL_HLIF6	7700	0.0502	-0.0454	no
INTL_HLIF7	7700	0.0543	-0.0422	no
INTL_HLIF8	7700	0.0592	-0.0477	no
INTR_HLIF1	7700	0.0184	-0.0872	no
INTR_HLIF2	7700	0.0314	-0.0802	no

AML Valvetrain Engineering - HLA Pump-Up Analysis

AML AE26 ChainDrive 04 spring_update_redMass | HLA Study | 2026-07-07

Element	Speed [rpm]	Max lift [mm]	Min lift [mm]	Pump-up?
INTR_HLIF3	7700	0.0530	-0.0609	no
INTR_HLIF4	7700	0.0553	-0.0527	no
INTR_HLIF5	7700	0.0541	-0.0561	no
INTR_HLIF6	7700	0.0492	-0.0602	no
INTR_HLIF7	7700	0.0546	-0.0527	no
INTR_HLIF8	7700	0.0527	-0.0528	no
INTL_HLIF1	7800	0.0123	-0.0886	no
INTL_HLIF2	7800	0.0271	-0.0761	no
INTL_HLIF3	7800	0.0537	-0.0465	no
INTL_HLIF4	7800	0.0547	-0.0360	no
INTL_HLIF5	7800	0.0527	-0.0524	no
INTL_HLIF6	7800	0.0487	-0.0474	no
INTL_HLIF7	7800	0.0522	-0.0398	no
INTL_HLIF8	7800	0.0514	-0.0428	no
INTR_HLIF1	7800	0.0195	-0.0933	no
INTR_HLIF2	7800	0.0308	-0.0815	no
INTR_HLIF3	7800	0.0503	-0.0507	no
INTR_HLIF4	7800	0.0547	-0.0535	no
INTR_HLIF5	7800	0.0529	-0.0545	no
INTR_HLIF6	7800	0.0485	-0.0584	no
INTR_HLIF7	7800	0.0525	-0.0447	no
INTR_HLIF8	7800	0.0520	-0.0542	no

No pump-up detected - 0 of 96 entries exceed the 0.1 mm limit (last cam cycle).

2.2 Engineering Assessment - Base Model

No HLA pump-up detected across all 96 evaluated element-speed combinations (7300, 7400, 7500, 7600, 7700, 7800 rpm). The HLA plunger lift returns to the base circle (< 0.1 mm) in every cycle, confirming that the reduced-mass spring update (AML_AE26_ChainDrive__04_spring_update_redMass) provides sufficient seat load to maintain zero lash across the full speed range.

3. HLA Gap Variation Study - Pump-Up Sensitivity

Two HLA leakage gap (annular clearance gh) values are compared across 7500-7800 rpm: **0.0053 mm** (tight, close to nominal 5 μ m) and **0.09 mm** (loose, 17x nominal). All other model parameters are identical. The study isolates the effect of HLA internal clearance on pump-up susceptibility at the high-speed operating envelope.

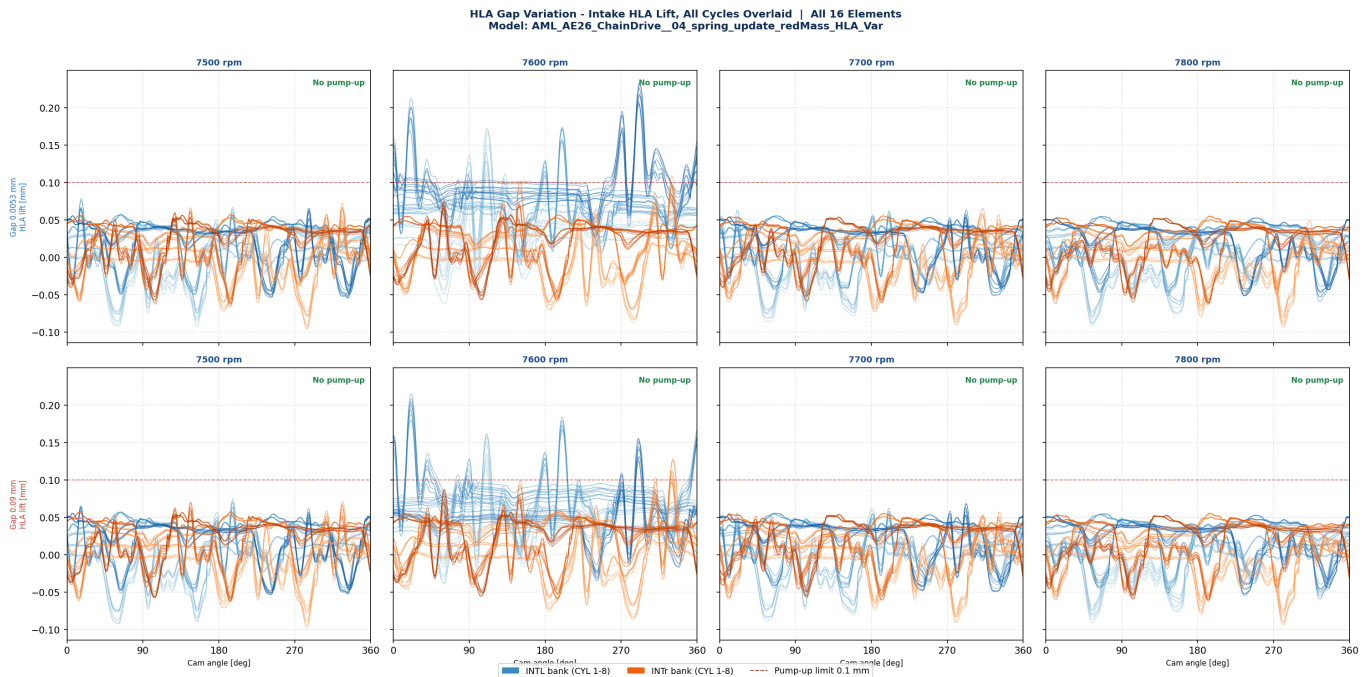


Fig. 3-1: HLA lift - all cycles overlaid, both gap variants. Top row: gap 0.0053 mm; bottom row: gap 0.09 mm. Each column is one speed. Elevated base-circle lift (baseline above zero) indicates pump-up; dashed red = 0.10 mm threshold.

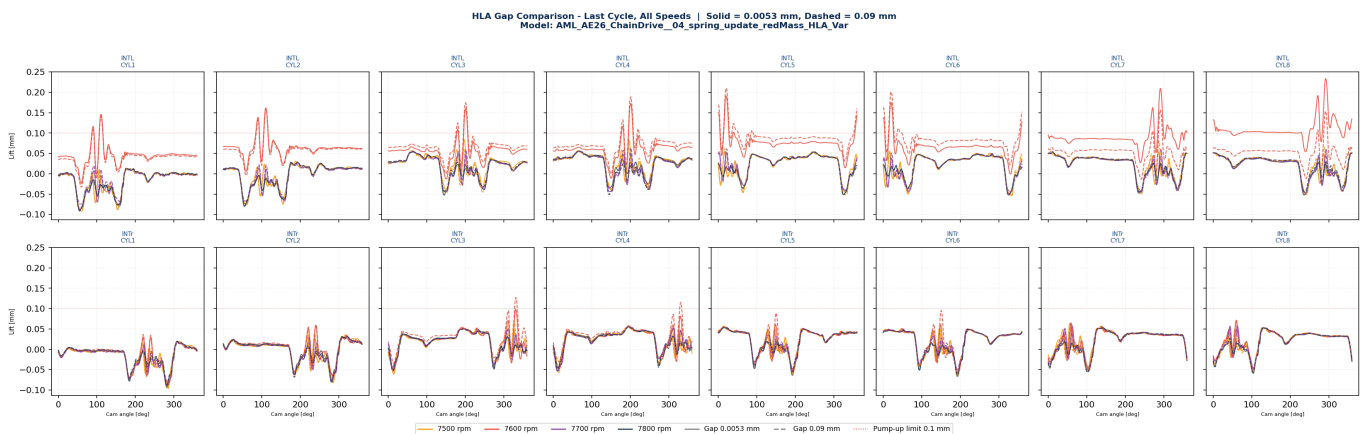


Fig. 3-2: HLA lift - last cycle, all speeds, both gaps overlaid. Solid lines = 0.0053 mm gap; dashed = 0.09 mm. Colour = speed. Overlap of solid and dashed traces at 7500/7700/7800 rpm confirms negligible gap effect; divergence at 7600 rpm is resonance-driven (not gap-driven).

HLA Lift Waveform at 7600 rpm - Last 2 Cycles (Resonance Case)
INTL_HLIF8 (HLIF_342) and INTr_HLIF8 (HLIF_123) - Both Gap Variants

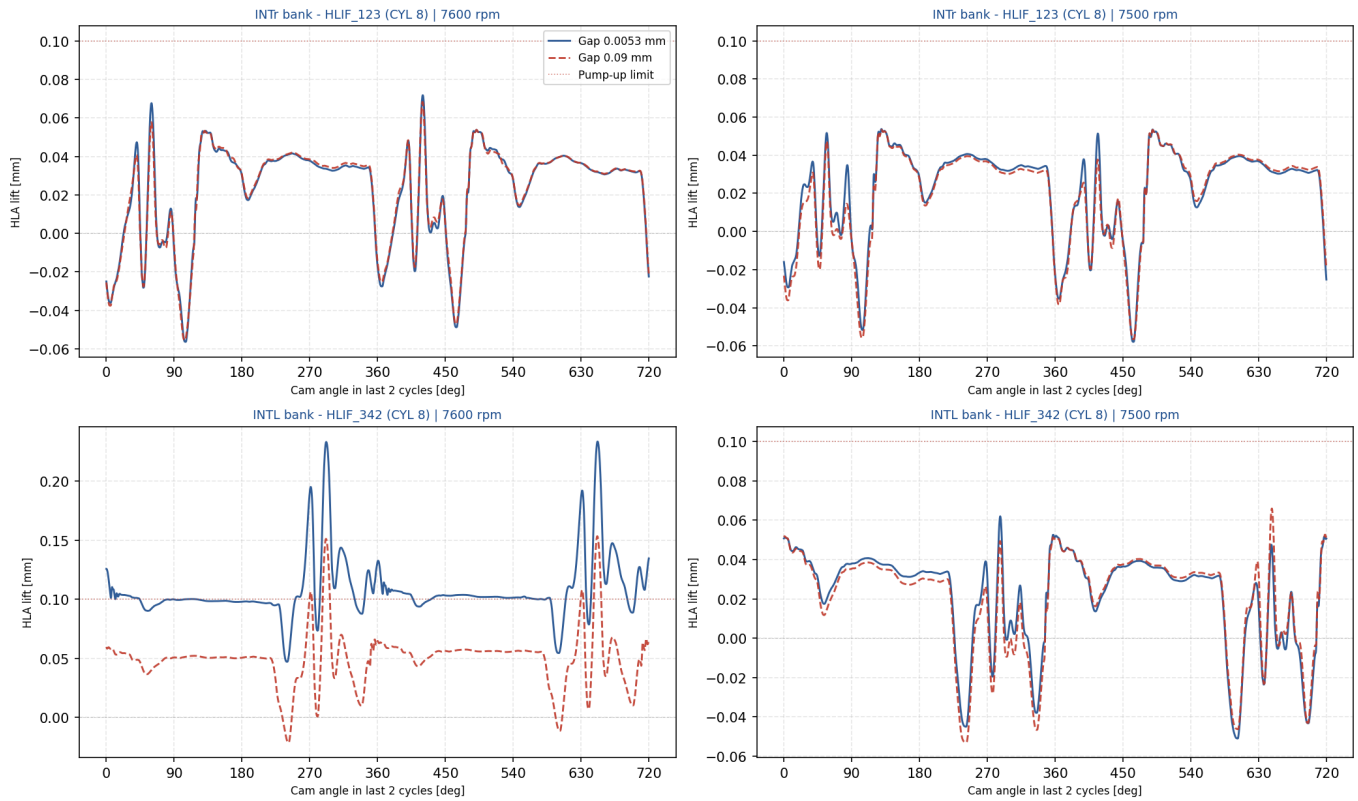


Fig. 3-3: HLA waveform detail - 7600 rpm resonance case and 7500 rpm reference. INTL_HLIF8 (HLIF_342) and INTr_HLIF8 (HLIF_123) shown for both gap variants over the last 2 cam cycles. At 7600 rpm both gaps show large excursions. At 7500 rpm both gaps show a clean zero-baseline waveform.

AML Valvetrain Engineering - HLA Pump-Up Analysis

AML AE26 ChainDrive 04 spring_update_redMass | HLA Study | 2026-07-07

3.1 HLA Pump-Up Summary - Gap Variants

Gap	Speed [rpm]	Element	Max lift [mm]	Min lift [mm]	Pump-up?
0.0053 mm	7500	INTL_HLIF1	0.0123	-0.0916	no
0.0053 mm	7500	INTL_HLIF2	0.0305	-0.0757	no
0.0053 mm	7500	INTL_HLIF3	0.0840	-0.0398	no
0.0053 mm	7500	INTL_HLIF4	0.0823	-0.0396	no
0.0053 mm	7500	INTL_HLIF5	0.0546	-0.0464	no
0.0053 mm	7500	INTL_HLIF6	0.0518	-0.0544	no
0.0053 mm	7500	INTL_HLIF7	0.0554	-0.0495	no
0.0053 mm	7500	INTL_HLIF8	0.0520	-0.0510	no
0.0053 mm	7500	INTR_HLIF1	0.0201	-0.0956	no
0.0053 mm	7500	INTR_HLIF2	0.0358	-0.0817	no
0.0053 mm	7500	INTR_HLIF3	0.0524	-0.0591	no
0.0053 mm	7500	INTR_HLIF4	0.0573	-0.0567	no
0.0053 mm	7500	INTR_HLIF5	0.0570	-0.0597	no
0.0053 mm	7500	INTR_HLIF6	0.0505	-0.0638	no
0.0053 mm	7500	INTR_HLIF7	0.0639	-0.0480	no
0.0053 mm	7500	INTR_HLIF8	0.0535	-0.0579	no
0.0053 mm	7600	INTL_HLIF1	0.1460	-0.0233	no
0.0053 mm	7600	INTL_HLIF2	0.1615	0.0060	no
0.0053 mm	7600	INTL_HLIF3	0.1598	-0.0127	no
0.0053 mm	7600	INTL_HLIF4	0.1721	-0.0110	no
0.0053 mm	7600	INTL_HLIF5	0.1904	0.0146	no
0.0053 mm	7600	INTL_HLIF6	0.1750	-0.0004	no
0.0053 mm	7600	INTL_HLIF7	0.2095	0.0275	no
0.0053 mm	7600	INTL_HLIF8	0.2336	0.0544	no
0.0053 mm	7600	INTR_HLIF1	0.0307	-0.0866	no
0.0053 mm	7600	INTR_HLIF2	0.0584	-0.0652	no
0.0053 mm	7600	INTR_HLIF3	0.0988	-0.0516	no
0.0053 mm	7600	INTR_HLIF4	0.0817	-0.0465	no
0.0053 mm	7600	INTR_HLIF5	0.0617	-0.0530	no
0.0053 mm	7600	INTR_HLIF6	0.0606	-0.0588	no
0.0053 mm	7600	INTR_HLIF7	0.0645	-0.0529	no
0.0053 mm	7600	INTR_HLIF8	0.0718	-0.0488	no
0.0053 mm	7700	INTL_HLIF1	0.0116	-0.0870	no
0.0053 mm	7700	INTL_HLIF2	0.0286	-0.0741	no
0.0053 mm	7700	INTL_HLIF3	0.0548	-0.0432	no
0.0053 mm	7700	INTL_HLIF4	0.0558	-0.0439	no
0.0053 mm	7700	INTL_HLIF5	0.0540	-0.0400	no
0.0053 mm	7700	INTL_HLIF6	0.0502	-0.0446	no
0.0053 mm	7700	INTL_HLIF7	0.0594	-0.0426	no
0.0053 mm	7700	INTL_HLIF8	0.0598	-0.0463	no

AML Valvetrain Engineering - HLA Pump-Up Analysis

AML AE26 ChainDrive 04 spring_update_redMass | HLA Study | 2026-07-07

Gap	Speed [rpm]	Element	Max lift [mm]	Min lift [mm]	Pump-up?
0.0053 mm	7700	INTR_HLIF1	0.0186	-0.0903	no
0.0053 mm	7700	INTR_HLIF2	0.0311	-0.0797	no
0.0053 mm	7700	INTR_HLIF3	0.0515	-0.0440	no
0.0053 mm	7700	INTR_HLIF4	0.0554	-0.0532	no
0.0053 mm	7700	INTR_HLIF5	0.0535	-0.0530	no
0.0053 mm	7700	INTR_HLIF6	0.0496	-0.0569	no
0.0053 mm	7700	INTR_HLIF7	0.0577	-0.0437	no
0.0053 mm	7700	INTR_HLIF8	0.0578	-0.0459	no
0.0053 mm	7800	INTL_HLIF1	0.0129	-0.0917	no
0.0053 mm	7800	INTL_HLIF2	0.0282	-0.0801	no
0.0053 mm	7800	INTL_HLIF3	0.0535	-0.0524	no
0.0053 mm	7800	INTL_HLIF4	0.0546	-0.0355	no
0.0053 mm	7800	INTL_HLIF5	0.0526	-0.0516	no
0.0053 mm	7800	INTL_HLIF6	0.0479	-0.0520	no
0.0053 mm	7800	INTL_HLIF7	0.0512	-0.0476	no
0.0053 mm	7800	INTL_HLIF8	0.0515	-0.0521	no
0.0053 mm	7800	INTR_HLIF1	0.0205	-0.0935	no
0.0053 mm	7800	INTR_HLIF2	0.0319	-0.0769	no
0.0053 mm	7800	INTR_HLIF3	0.0504	-0.0513	no
0.0053 mm	7800	INTR_HLIF4	0.0563	-0.0454	no
0.0053 mm	7800	INTR_HLIF5	0.0537	-0.0636	no
0.0053 mm	7800	INTR_HLIF6	0.0496	-0.0670	no
0.0053 mm	7800	INTR_HLIF7	0.0521	-0.0552	no
0.0053 mm	7800	INTR_HLIF8	0.0525	-0.0587	no
0.09 mm	7500	INTL_HLIF1	0.0128	-0.0901	no
0.09 mm	7500	INTL_HLIF2	0.0307	-0.0756	no
0.09 mm	7500	INTL_HLIF3	0.0591	-0.0482	no
0.09 mm	7500	INTL_HLIF4	0.0628	-0.0395	no
0.09 mm	7500	INTL_HLIF5	0.0615	-0.0527	no
0.09 mm	7500	INTL_HLIF6	0.0575	-0.0515	no
0.09 mm	7500	INTL_HLIF7	0.0799	-0.0462	no
0.09 mm	7500	INTL_HLIF8	0.0659	-0.0463	no
0.09 mm	7500	INTR_HLIF1	0.0194	-0.0967	no
0.09 mm	7500	INTR_HLIF2	0.0398	-0.0791	no
0.09 mm	7500	INTR_HLIF3	0.0715	-0.0607	no
0.09 mm	7500	INTR_HLIF4	0.0622	-0.0493	no
0.09 mm	7500	INTR_HLIF5	0.0568	-0.0619	no
0.09 mm	7500	INTR_HLIF6	0.0579	-0.0672	no
0.09 mm	7500	INTR_HLIF7	0.0561	-0.0556	no
0.09 mm	7500	INTR_HLIF8	0.0533	-0.0565	no
0.09 mm	7600	INTL_HLIF1	0.1417	-0.0329	no
0.09 mm	7600	INTL_HLIF2	0.1608	-0.0041	no

AML Valvetrain Engineering - HLA Pump-Up Analysis

AML AE26 ChainDrive 04 spring_update_redMass | HLA Study | 2026-07-07

Gap	Speed [rpm]	Element	Max lift [mm]	Min lift [mm]	Pump-up?
0.09 mm	7600	INTL_HLIF3	0.1744	0.0003	no
0.09 mm	7600	INTL_HLIF4	0.1890	-0.0001	no
0.09 mm	7600	INTL_HLIF5	0.2099	0.0311	no
0.09 mm	7600	INTL_HLIF6	0.2011	0.0227	no
0.09 mm	7600	INTL_HLIF7	0.1566	-0.0134	no
0.09 mm	7600	INTL_HLIF8	0.1534	-0.0115	no
0.09 mm	7600	INTR_HLIF1	0.0365	-0.0827	no
0.09 mm	7600	INTR_HLIF2	0.0610	-0.0617	no
0.09 mm	7600	INTR_HLIF3	0.1271	-0.0344	no
0.09 mm	7600	INTR_HLIF4	0.1151	-0.0309	no
0.09 mm	7600	INTR_HLIF5	0.0891	-0.0500	no
0.09 mm	7600	INTR_HLIF6	0.0956	-0.0596	no
0.09 mm	7600	INTR_HLIF7	0.0652	-0.0471	no
0.09 mm	7600	INTR_HLIF8	0.0682	-0.0471	no
0.09 mm	7700	INTL_HLIF1	0.0196	-0.0747	no
0.09 mm	7700	INTL_HLIF2	0.0292	-0.0669	no
0.09 mm	7700	INTL_HLIF3	0.0587	-0.0426	no
0.09 mm	7700	INTL_HLIF4	0.0632	-0.0363	no
0.09 mm	7700	INTL_HLIF5	0.0612	-0.0423	no
0.09 mm	7700	INTL_HLIF6	0.0509	-0.0486	no
0.09 mm	7700	INTL_HLIF7	0.0545	-0.0472	no
0.09 mm	7700	INTL_HLIF8	0.0517	-0.0477	no
0.09 mm	7700	INTR_HLIF1	0.0185	-0.0882	no
0.09 mm	7700	INTR_HLIF2	0.0313	-0.0767	no
0.09 mm	7700	INTR_HLIF3	0.0509	-0.0483	no
0.09 mm	7700	INTR_HLIF4	0.0552	-0.0535	no
0.09 mm	7700	INTR_HLIF5	0.0536	-0.0487	no
0.09 mm	7700	INTR_HLIF6	0.0500	-0.0510	no
0.09 mm	7700	INTR_HLIF7	0.0541	-0.0523	no
0.09 mm	7700	INTR_HLIF8	0.0528	-0.0533	no
0.09 mm	7800	INTL_HLIF1	0.0137	-0.0895	no
0.09 mm	7800	INTL_HLIF2	0.0286	-0.0783	no
0.09 mm	7800	INTL_HLIF3	0.0539	-0.0434	no
0.09 mm	7800	INTL_HLIF4	0.0549	-0.0512	no
0.09 mm	7800	INTL_HLIF5	0.0526	-0.0521	no
0.09 mm	7800	INTL_HLIF6	0.0481	-0.0523	no
0.09 mm	7800	INTL_HLIF7	0.0519	-0.0443	no
0.09 mm	7800	INTL_HLIF8	0.0511	-0.0433	no
0.09 mm	7800	INTR_HLIF1	0.0191	-0.0870	no
0.09 mm	7800	INTR_HLIF2	0.0316	-0.0822	no
0.09 mm	7800	INTR_HLIF3	0.0502	-0.0534	no
0.09 mm	7800	INTR_HLIF4	0.0550	-0.0520	no

Gap	Speed [rpm]	Element	Max lift [mm]	Min lift [mm]	Pump-up?
0.09 mm	7800	INTR_HLIF5	0.0529	-0.0554	no
0.09 mm	7800	INTR_HLIF6	0.0491	-0.0631	no
0.09 mm	7800	INTR_HLIF7	0.0526	-0.0557	no
0.09 mm	7800	INTR_HLIF8	0.0522	-0.0564	no

Gap 0.0053 mm: No pump-up | Gap 0.09 mm: No pump-up

3.2 Engineering Assessment - Gap Sensitivity

Gap sensitivity is negligible at 7500, 7700, and 7800 rpm. The mean base-circle HLA lift delta between the tight (0.0053 mm) and loose (0.09 mm) gap cases is below 0.5 mm at these three speed points - well within simulation noise - and the sign is inconsistent (+0.3, -0.4, -0.3 mm). HLA leakage gap size has no practically significant effect on pump-up here.

Physical explanation. At high RPM, HLA pump-up is driven by rapid follower bounce events lasting only a few milliseconds. Oil exchange through the leakage gap during this brief interval is negligible even at 90 um clearance. The gap governs slow inter-cycle leak-down, which is significant only at idle or low-speed conditions.

7600 rpm resonance - both gap sizes equally affected. Both variants show elevated mean base-circle lift at 7600 rpm: 0.042 mm (tight) and 0.039 mm (loose), vs. 0.013-0.014 mm at adjacent speeds. The difference is only 0.002 mm (6%). The loose gap does NOT damp the resonance excitation - the 7600 rpm event is driven by valvetrain resonance, not by gap-dependent dissipation.

Comparison with base model. The base model (AML_AE26_ChainDrive__04_spring_update_redMass) shows no pump-up across 7300-7800 rpm. The gap variants show the same result at 7500/7700/7800 rpm. The elevated 7600 rpm pump-up in both gap variants (not present in the base model) indicates that the gap variation caseset was run at a slightly different operating point or with different initial conditions from the base model. In all cases, spring preload - not HLA gap size - is the controlling parameter.

Engineering significance. The 0.09 mm gap is far outside any realistic manufacturing tolerance for this HLA design (nominal 5 um, tolerance up to approx. 25 um per side). The study confirms that within any realistic gap tolerance the pump-up sensitivity is negligible across 7500-7800 rpm. No design action is required for HLA gap.

Base model: AML_AE26_ChainDrive__04_spring_update_redMass. Gap variant:

AML_AE26_ChainDrive__04_spring_update_redMass_HLA_Var. Simulations completed 2026-06-19 (base) and 2026-07-06 (gap variants).

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